

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (IT) Examination****March-April 2018****IT314 Operating System Fundamentals****Date: 27.03.2018, Tuesday****Time: 1:30 p.m. To 4:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION I**Q – 1.a Answer the following questions:**

1. What do you mean by mode switch? **[01]**
2. Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end. **[02]**
3. What is a Semaphore? **[01]**
4. Define the following terms: **[02]**
i) Degree of Multiprogramming ii) Swapping
5. An edge from process P_i to P_j in a wait for graph indicates that : **[01]**
i) P_i is waiting for P_j to release a resource that P_i needs
ii) P_j is waiting for P_i to release a resource that P_j needs
iii) P_i is waiting for P_j to leave the system
iv) P_j is waiting for P_i to leave the system

Q – 1.b Draw and explain the 5 State Process transition diagram. [05]

Q – 1.c A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below. Find whether system is in safe state or nor? **[05]**
 If yes, then give the safe sequence.

Process	Current Allocation	Maximum Requirement
P1	3	7
P2	1	6
P3	3	5

Q – 2. Answer the following:(Any Two)**[08]**

- a. Give the difference between User Level Thread and Kernel Level Thread.
- b. What is Race Condition? How do you avoid it?
- c. Assume that the following jobs are to be executed on a single processor system

Job ID	CPU Burst Time(milli seconds)
P	4
Q	1
R	8
S	1
T	2

The jobs are assumed to have arrived at time 0 and in the order p, q, r, s, t. Draw the Gantt chart and Calculate the average turnaround time for Round Robin Algorithm(time quantum=1)

- d. Consider the following set of processes, with the length of the CPU burst and arrival time given in milliseconds:

Process	Arrival Time	Burst Time
P1	0.0	8
P2	0.4	4
P3	1.0	1

Draw Gantt chart that illustrates the execution of these processes using First Come First Serve Scheduling (FCFS) Algorithm. Find the average waiting time for FCFS.

Q - 3 Answer the following:(Any Two)**[10]**

- a. Define operating System. How does Operating system do time sharing? What are the advantages of time sharing?
- b. Explain Monolithic and Microkernel approach of operating system.
- c. Discuss in detail Producer Consumer Problem in Inter Process Communication.

SECTION – II

Q – 4.a Answer the following questions:

1. Define Thrashing. How can you minimize thrashing? [02]
2. What is a device driver? [01]
3. Explain Internal Fragmentation. [02]
4. Airline reservation systems and inventory control system are the examples of system. [01]
i) pile ii) sequential file iii) indexed sequential file iv) indexed file
5. What is mounting a file system? [01]
i) Creating a file system
ii) Deleting a file system
iii) Attaching portion of file into directory structure
iv) Removing portion of file from directory structure

Q – 4.b A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string): 1, 2, 1, 3, 7, 4, 5, 6, 3, 1. Find the total number of page faults when using following page replacement algorithm [05]
(a) Optimal (b) FCFS

Q – 4.c Given five memory partitions of 100kb, 500kb, 200kb, 300kb and 600 kb (in order), how would the first fit and best fit place processes of 212 kb, 417 kb, 112 kb and 426 kb (in order)? Which algorithm makes most efficient use of memory? [05]

Q – 5 Answer the following questions: (Any Two) [08]

- a. What are the four necessary conditions for deadlock?
- b. A disk has 200 cylinders. Request comes in following order. The arm is initially at 50. 95, 180, 34, 119, 11, 123, 62, 64
Find the disk arm movement in case of
(i) FCFS (ii) SSTF
- c. Explain File Locking mechanism in File Management.
- d. How does Direct access file method differ from Indexed Sequential File access method?

Q – 6 Answer the following questions: (Any Two) [10]

- a. How does Worm and Trojan Horse affect our computer? Which are the program threats posed by them?
- b. Write a Short Note on: Direct Memory Access.
- c. What is a Page Table? How does page table map logical address to physical address?
